

Query Bootloader Version	Command Packet Format	0x03	0x00	Index_id	0xF9	CRC						
	Response Packet Format	0x04	0x00	Index_id	0xF9	Bootloader_version	CRC					
	Command Details	(1) Function: Query the bootloader version. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Bootloader_version (4) Command code: 0xF9										
Query Device Firmware Version	Command Packet Format	0x03	0x00	Index_id	0xFA	CRC						
	Response Packet Format	0x04	0x00	Index_id	0xFA	Firmware_version	CRC					
	Command Details	(1) Function: Query the device firmware version. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Firmware_version (4) Command code: 0xFA										
Query Device Type	Command Packet Format	0x03	0x00	Index_id	0xFB	CRC						
	Response Packet Format	0x05	0x00	Index_id	0xFB	Device_type_low	Device_type_high	CRC				
	Command Details	(1) Function: Query the device type. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Device_type (4) Command code: 0xFB Note 1: Device_type = (uint16_t)((Device_type_high << 8) Device_type_low)										
Enumeration Request	Command Packet Format	None										
	Response Packet Format	0x03	0x00	Index_id	0xFC	CRC						
	Command Details	(1) Function: Enumeration request. Sent by the terminal device when the chain link changes and sent when the device powers on, notifying the host to update the chain device status. (2) Input parameter: none (3) Return parameter: none (4) Command code: 0xFC										
Heartbeat Packet	Command Packet Format	0x03	0x00	0xFF	0xFD	CRC						
	Response Packet Format	0x03	0x00	0xFF	0xFD	CRC						
	Command Details	(1) Function: Heartbeat packet. Chain devices communicate periodically to determine whether they are the terminal device. The host can also use heartbeat packets to determine whether chain devices are connected. (2) Input parameter: none (3) Return parameter: none (4) Command code: 0xFD										
Enumerate	Command Packet Format	0x04	0x00	0xFF	0xFE	Send_num	CRC					
	Response Packet Format	0x04	0x00	0xFF	0xFE	Receive_num	CRC					
	Command Details	(1) Function: The user can obtain the number of devices in the chain through enumeration. (2) Input parameter: Send_num (default: 0, used to record the number of devices) (3) Return parameter: Receive_num (the value represents the number of devices) (4) Command code: 0xFE										
Note 1: The maximum packet length is 256 bytes for serial communication Note 2: The data length is from Index_id to CRC, including Index_id and CRC, but excluding the data length field itself Note 3: During CRC calculation, exclude the packet header, packet tail, length, and CRC fields themselves, and sum only the remaining data Note 4: Serial communication baud rate: 115200; 8 data bits; 1 stop bit; no parity bit <pre>uint8_t calculateCRC(const uint8_t *buffer, uint16_t size) { uint8_t crc8 = 0; for (uint8_t i = 4; i < (size - 3); i++) { crc8 += buffer[i]; } return crc8; }</pre>												